① The time required for 50% consolidation of a 25-mm-thick clay layer (drained at both top and bottom) in the laboratory is 2 min 20 sec

a) How long (in days) it take for 3m thick clay layer of the same clay in the field under the same pressure increment to reach 50% consolidation? In the field, sand layers are present at the top and bottom of the clay layer

b) How long will it take in the field for 30% primary Consolidation occur?

Solution

a) $U = 50\% \Rightarrow T_v = 0.197$

$$C_v = \frac{T_v \cdot (H_{dr})^2}{t}, \quad H_{dr} = \frac{0.025}{2} = 0.0125 \text{ cm}$$

$$C_v = \frac{0.197 \cdot (0.0125)^2}{140} \Rightarrow C_v = 2.198 \times 10^{-7} \text{ m}^2/\text{sec}.$$

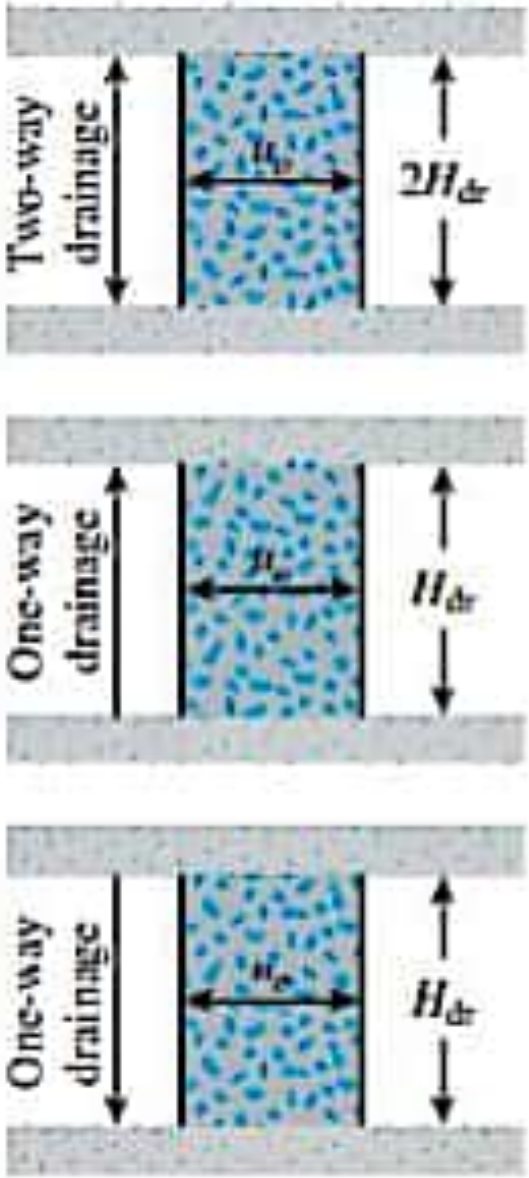
$$2.198 \times 10^{-7} = \frac{0.197 \cdot \left(\frac{3}{2}\right)^2}{t} \Rightarrow t = 2016606.5 \text{ sec} = 23.34 \text{ days}$$

b) $U = 30\% \Rightarrow T_v = 0.0707$

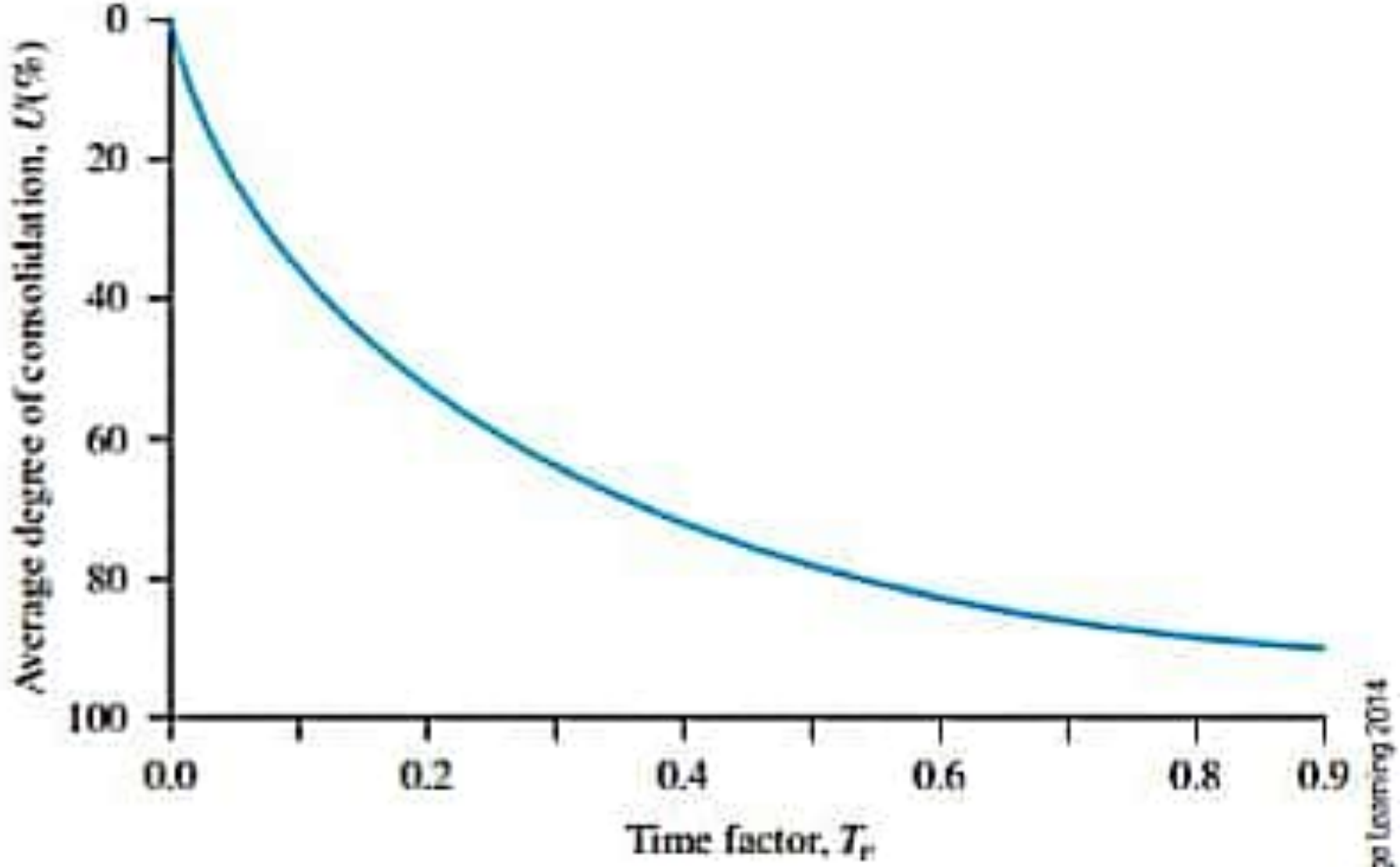
$$C_v = \frac{T_v \cdot (H_{dr})^2}{t}$$

$$2.198 \times 10^{-7} = \frac{0.0707 \cdot \left(\frac{3}{2}\right)^2}{t}$$

$$t = 8.37 \text{ days}$$



Different types of drainage with u_o constant



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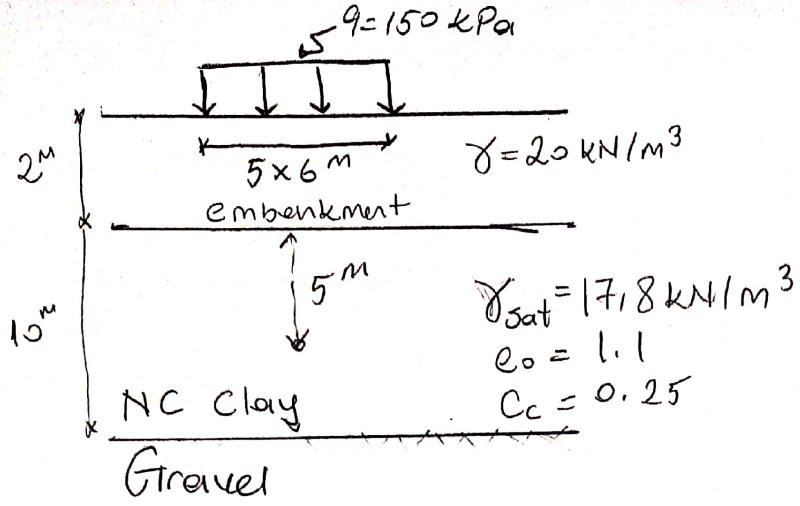
Figure 11.26 Variation of average degree of consolidation with time factor, T_r (u_o constant with depth)

Table 11.7 Variation of T_v with U

U (%)	T_v	U (%)	T_v	U (%)	T_v	U (%)	T_v
0	0	26	0.0531	52	0.212	78	0.529
1	0.00008	27	0.0572	53	0.221	79	0.547
2	0.0003	28	0.0615	54	0.230	80	0.567
3	0.00071	29	0.0660	55	0.239	81	0.588
4	0.00126	30	0.0707	56	0.248	82	0.610
5	0.00196	31	0.0754	57	0.257	83	0.633
6	0.00283	32	0.0803	58	0.267	84	0.658
7	0.00385	33	0.0855	59	0.276	85	0.684
8	0.00502	34	0.0907	60	0.286	86	0.712
9	0.00636	35	0.0962	61	0.297	87	0.742
10	0.00785	36	0.102	62	0.307	88	0.774
11	0.0095	37	0.107	63	0.318	89	0.809
12	0.0113	38	0.113	64	0.329	90	0.848
13	0.0133	39	0.119	65	0.340	91	0.891
14	0.0154	40	0.126	66	0.352	92	0.938
15	0.0177	41	0.132	67	0.364	93	0.993
16	0.0201	42	0.138	68	0.377	94	1.055
17	0.0227	43	0.145	69	0.390	95	1.129
18	0.0254	44	0.152	70	0.403	96	1.219
19	0.0283	45	0.159	71	0.417	97	1.336
20	0.0314	46	0.166	72	0.431	98	1.500
21	0.0346	47	0.173	73	0.446	99	1.781
22	0.0380	48	0.181	74	0.461	100	∞
23	0.0415	49	0.188	75	0.477		
24	0.0452	50	0.197	76	0.493		
25	0.0491	51	0.204	77	0.511		

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(2)



a) What is the consolidation settlement?
- use 2:1 method
- assume $\Delta\sigma_1 = 150$ kPa

b) The time required for 50% consolidation of a 25 mm thick clay layer in laboratory 12 min. How long will it take for a 10 m thick clay layer in the field to reach 50% consolidation.

Solution

a $S_c = \frac{C_c}{1+e_0} \cdot H \cdot \log\left(\frac{\sigma_{v0}' + \Delta\sigma'}{\sigma_{v0}'}\right)$

$\sigma_{v0}' = 2\times 20 + 5\times(17.8 - 9.81) = 80$ kPa

→ Assume $\Delta\sigma' = 150$ kPa.

$$S_c = \frac{0.25}{1+1.1} \cdot 10 \cdot \log\left(\frac{80+150}{80}\right) = 0.55\ m$$

* Use simplified method for calculation of stress increase ($\Delta\sigma$) in the middle of clay layer

$z = 2+5 = 7^m \Rightarrow \Delta\sigma = \frac{150\times 5\times 6}{(5+7)(6+7)} = 28.85$ kPa

$$S_c = \frac{0.25}{1+1.1} \cdot 10 \cdot \log\left(\frac{80+28.85}{80}\right) = 0.16\ m$$

b) $U = 50\%$ $T_v = 0.197$

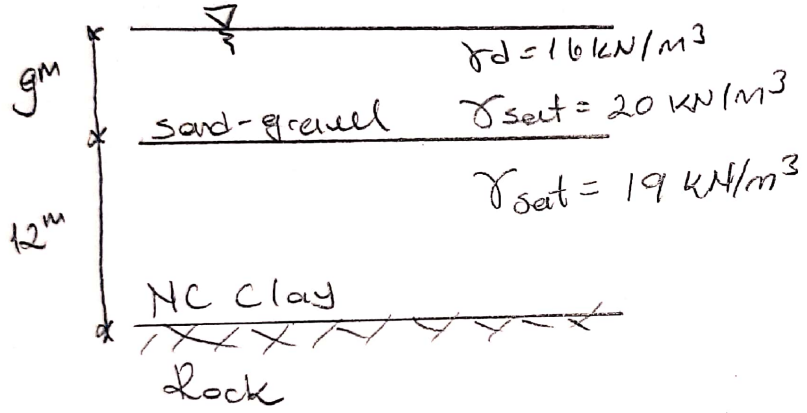
* In laboratory, the time required for 50% consolidation is $t = 12$ min. The drainage of water is possible for both top and bottom side of the sample.

$$C_v = \frac{T_v \cdot H_d^2}{t}, \quad C_v = \frac{0.197 \times \left(\frac{2.5}{2}\right)^2}{12} \Rightarrow C_v = 0.0256 \text{ cm}^2/\text{min}$$

$$C_v = \frac{T_v \cdot H_d^2}{t}, \quad 0.0256 = \frac{0.197 \times \left(\frac{1000}{2}\right)^2}{t} \Rightarrow t = 3.66 \text{ years}$$

$(1920077.973 \text{ min})$

(3)



σ (kPa)	e
0	0.90
25	0.89
50	0.87
100	0.85
200	0.814
400	0.775
800	0.736

a) what is the consolidation settlement if ground water level is lowered 3m

b) $C_v = 2.59 \times 10^{-6}$ m²/sec what is the consolidation settlement after 1 year

c) $m_v = 0.00019$ m²/kN, $S_c = ?$

Solution

1st situation → $\sigma'_{v_{o_1}} = 9 \times (20 - 9.81) + 6 \times (19 - 9.81) = 146.85$ kPa.

2nd situation → $\sigma'_{v_{o_2}} = 3 \times 16 + 6 \times (20 - 9.81) + 6 \times (19 - 9.81) = 164.28$ kPa

$\left\{ \sigma'_{v_{o_2}} - \sigma'_{v_{o_1}} = 17.43 \text{ kPa} = \Delta\sigma_v \right\}$ → Due to the drop of water level, the effective stress will increase

$$C_c = \frac{0.814 - 0.775}{\log(400/200)} = 0.129 \text{ (for 200 kPa and 400 kPa)}$$

$$S_c = \frac{c_c}{1+e_o} \cdot H \cdot \log\left(\frac{\sigma_{vo}' + \Delta\sigma'}{\sigma_{vo}'}\right)$$

$$S_c = \frac{0.129}{1+0.9} \cdot 12 \cdot \log\left(\frac{146.85 + 17.43}{146.85}\right)$$

$S_c = 4\ cm$

b) $c_v = \frac{T_v \cdot H_{dr}^2}{t}$, $2.59 \times 10^{-6} = \frac{T_v \times (12)^2}{31536000}$ ← Rock → one drainage way.

$c_v = 2.59 \times 10^{-6}\ m^2/sec$

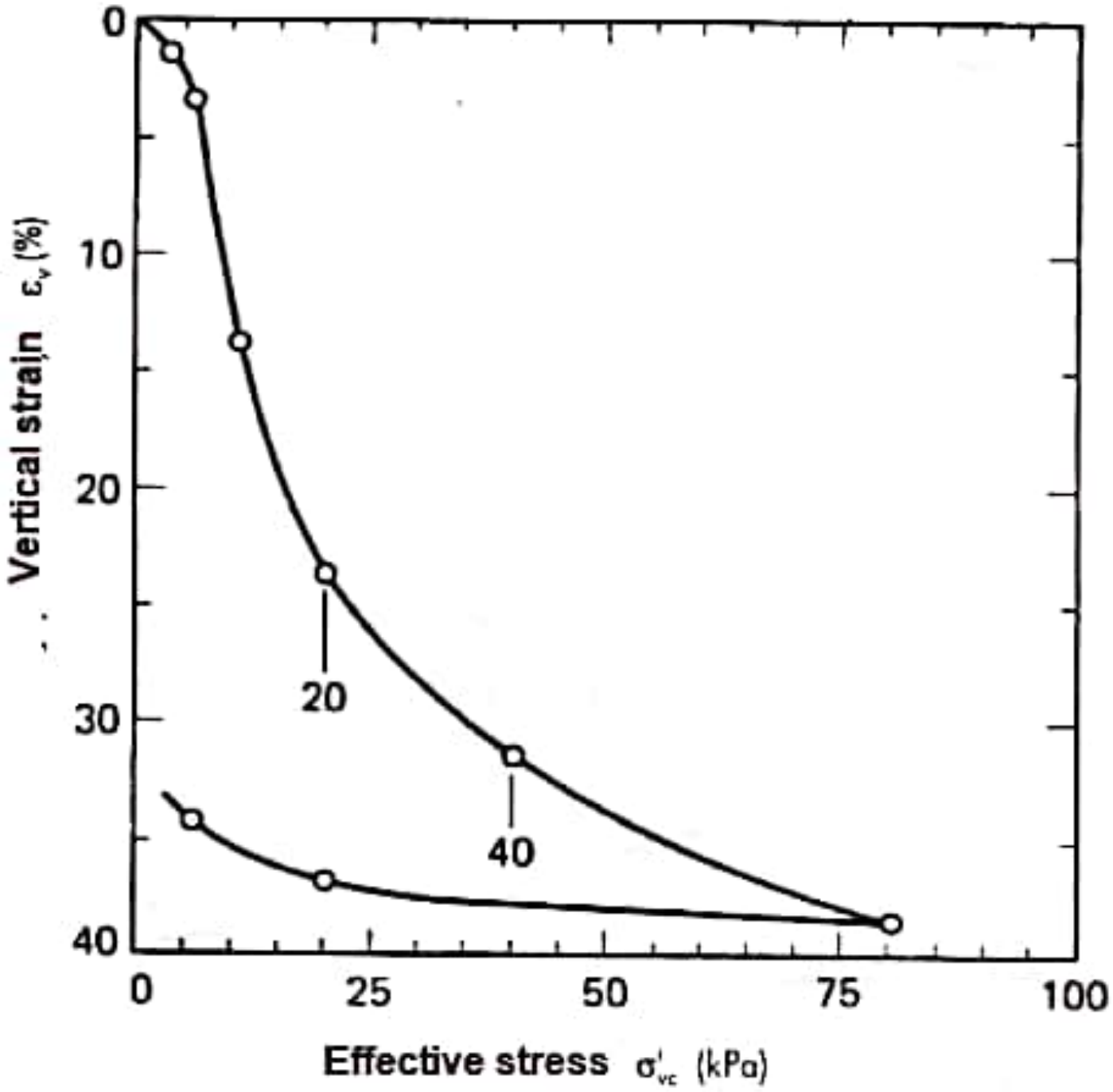
$T_v = 0.567 \rightarrow U = 80\%$

24 × 80% → 3.2 cm settlement in one year

C) m_v = coefficient of volumetric compressibility

$S_c = m_v \cdot H \cdot \Delta\sigma'$

$S_c = 0.00019 \times 12 \times 12 = 2.73\ cm$



Coefficient of volumetric compressibility

$$m_v = \frac{\Delta \varepsilon_v}{\Delta \sigma'}$$

$$m_v = \frac{\Delta H / H_0}{\Delta \sigma'}$$

$$m_v = \frac{a_v}{1 + e_0}$$

$$m_v = \frac{\Delta H / H_0}{\Delta \sigma'}$$



$$\Delta H = S_\infty = H_0 . m_v . \Delta \sigma'$$

(total settlement)

The value of mv for a particular soil is not constant but depends on the stress range over which it is calculated.